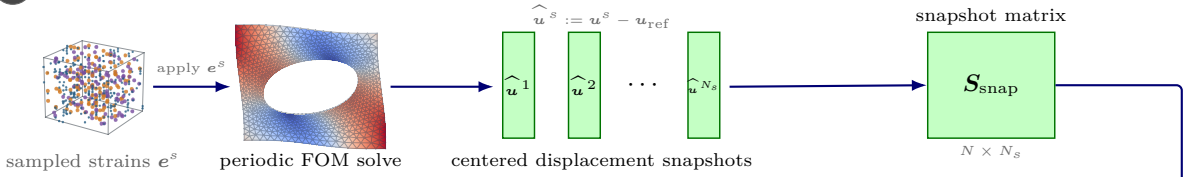


1 RVE snapshot generation

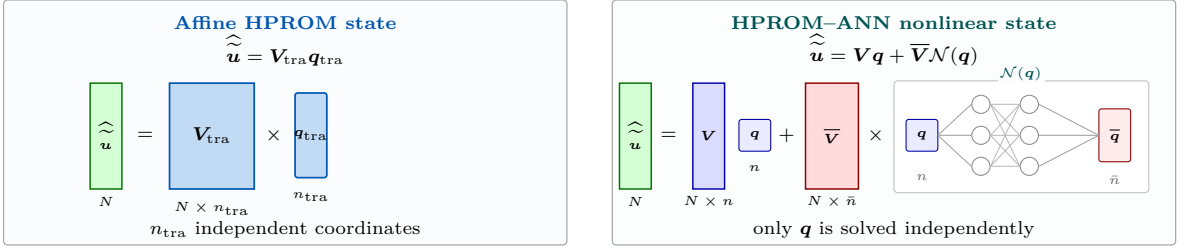


2 From the snapshot matrix to the reduced bases



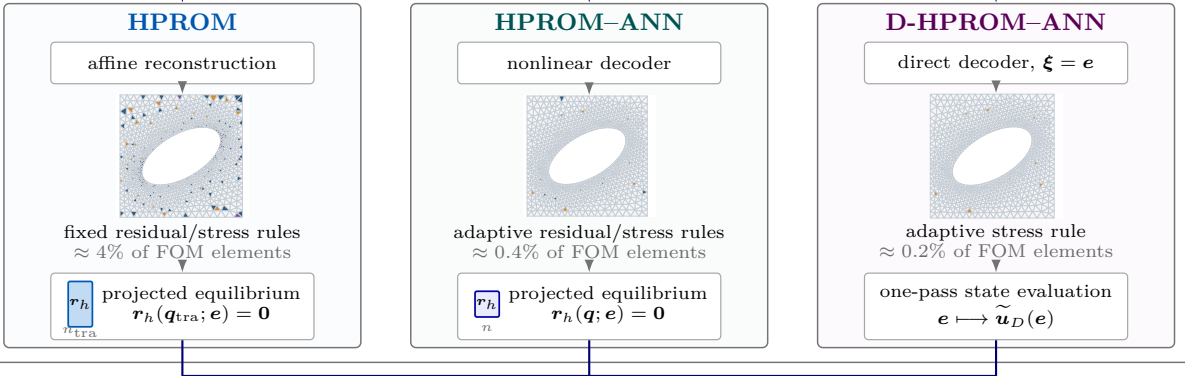
3 Two reduced representations of the microscopic state

$$\hat{u} := \tilde{u} - u_{\text{ref}}$$



4 Three online routes

prescribed macroscopic strain e



5 Common constitutive output



same FE² interface; different state, integration, and equilibrium approximations